

Commerce XI Maths App

Theory (Pages 38–40)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 20

Question 1 - Theory P38

To find $\sqrt{x + iy}$, the textbook assumes

Options

- A. $\sqrt{x + iy} = a + ib$, where $a, b \in \mathbb{R}$
- B. $\sqrt{x + iy} = a + b$
- C. $\sqrt{x + iy} = ai + b$
- D. $\sqrt{x + iy} = x + iy$

Correct answer: $\sqrt{x + iy} = a + ib$, where $a, b \in \mathbb{R}$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\sqrt{x + iy} = a + ib$, where $a, b \in \mathbb{R}$.

Question 2 - Theory P38

If $x + iy = (a + ib)^2$, then x equals

Options

- A. $a^2 - b^2$
- B. $a^2 + b^2$
- C. $2ab$
- D. $a - b$

Correct answer: $a^2 - b^2$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $a^2 - b^2$.

Question 3 - Theory P38

If $x + iy = (a + ib)^2$, then y equals

Options

- A. $2ab$
- B. $a^2 - b^2$
- C. $a^2 + b^2$
- D. $a + b$

Correct answer: $2ab$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $2ab$.

Question 4 - Solved Example P38

The square roots of $6 + 8i$ are

Options

- A. $\pm\sqrt{2}(2 + i)$
- B. $\pm\sqrt{2}(2 - i)$
- C. $\pm(2 + 2i)$
- D. $\pm(\sqrt{6} + \sqrt{8}i)$

Correct answer: $\pm\sqrt{2}(2 + i)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\pm\sqrt{2}(2 + i)$.

Question 5 - Solved Example P38

One square root of $6 + 8i$ is

Options

- A. $2\sqrt{2} + \sqrt{2}i$
- B. $2\sqrt{2} - \sqrt{2}i$
- C. $\sqrt{2} + 2\sqrt{2}i$
- D. $-2\sqrt{2} + \sqrt{2}i$

Correct answer: $2\sqrt{2} + \sqrt{2}i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $2\sqrt{2} + \sqrt{2}i$.

Question 6 - Solved Example P39

The square roots of $2i$ are

Options

- A. $\pm(1 + i)$
- B. $\pm(1 - i)$
- C. $\pm 2i$
- D. $\pm\sqrt{2}i$

Correct answer: $\pm(1 + i)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\pm(1 + i)$.

Question 7 - Solved Example P39

In finding $\sqrt{(2)}i = a + ib$, equating parts gives

Options

- A. $a^2 - b^2 = 0$ and $2ab = 2$
- B. $a^2 + b^2 = 0$ and $ab = 2$
- C. $a^2 - b^2 = 2$ and $ab = 0$
- D. $a + b = 2$ and $ab = 1$

Correct answer: $a^2 - b^2 = 0$ and $2ab = 2$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $a^2 - b^2 = 0$ and $2ab = 2$.

Question 8 - Theory P39

The roots of $ax^2 + bx + c = 0$ are

Options

- A. $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$
- B. $(b \pm \sqrt{(b^2 + 4ac)})/(2a)$
- C. $(-b \pm \sqrt{(b^2 + 4ac)})/(a)$
- D. $(-b \pm \sqrt{(b^2 - ac)})/(2a)$

Correct answer: $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$.

Question 9 - Theory P39

The expression $b^2 - 4ac$ is called

Options

- A. the discriminant
- B. the conjugate
- C. the imaginary part
- D. the modulus

Correct answer: the discriminant

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives the discriminant.

Question 10 - Theory P39

If $a, b, c \in \mathbb{R}$ and $b^2 - 4ac < 0$, the roots are

Options

- A. complex numbers
- B. equal real numbers
- C. positive real numbers
- D. natural numbers

Correct answer: complex numbers

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives complex numbers.

Question 11 - Let's Note P39

If $p + iq$ is a root of a real-coefficient quadratic equation, the other root is

Options

- A. $p - iq$
- B. $-p + iq$
- C. $-p - iq$
- D. $q + ip$

Correct answer: $p - iq$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $p - iq$.

Question 12 - Let's Note P39

Complex roots of a real-coefficient quadratic equation occur in

Options

- A. conjugate pairs
- B. equal pairs only
- C. reciprocal pairs
- D. negative pairs

Correct answer: conjugate pairs

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives conjugate pairs.

Question 13 - Solved Example P39

Solve $x^2 + x + 1 = 0$

Options

A. $x = (-1 \pm \sqrt{3}) i / (2)$

B. $x = (1 \pm \sqrt{3}) i / (2)$

C. $x = -1 \pm \sqrt{3} i$

D. $x = (-1 \pm 3i) / (2)$

Correct answer: $x = (-1 \pm \sqrt{3}) i / (2)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $x = (-1 \pm \sqrt{3}) i / (2)$.

Question 14 - Solved Example P39

For $x^2 + x + 1 = 0$, the discriminant is

Options

A. -3

B. 3

C. -4

D. 5

Correct answer: -3

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -3.

Question 15 - Solved Example P39

Solve $x^2 - 4x + 13 = 0$

Options

A. $x = 2 \pm 3i$

B. $x = -2 \pm 3i$

C. $x = 4 \pm 13i$

D. $x = 2 \pm i$

Correct answer: $x = 2 \pm 3i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $x = 2 \pm 3i$.

Question 16 - Solved Example P39

For $x^2 - 4x + 13 = 0$, completing the square gives

Options

- A. $(x - 2)^2 = -9$
- B. $(x + 2)^2 = -9$
- C. $(x - 4)^2 = -13$
- D. $(x - 2)^2 = 9$

Correct answer: $(x - 2)^2 = -9$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(x - 2)^2 = -9$.

Question 17 - Solved Example P40

For $x^2 + 4ix - 5 = 0$, $b^2 - 4ac$ equals

Options

- A. 4
- B. -4
- C. 16
- D. -36

Correct answer: 4

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives 4.

Question 18 - Solved Example P40

Solve $x^2 + 4ix - 5 = 0$

Options

- A. $x = -2i \pm 1$
- B. $x = 2i \pm 1$
- C. $x = -4i \pm 5$
- D. $x = -2 \pm i$

Correct answer: $x = -2i \pm 1$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $x = -2i \pm 1$.

Question 19 - Theory P39

For a quadratic equation, the coefficient a must satisfy

Options

- A. $a \neq 0$
- B. $a = 0$
- C. $a > 0$ only
- D. $a = 1$ only

Correct answer: $a \neq 0$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $a \neq 0$.

Question 20 - Theory P39

If $D = b^2 - 4ac$, then $\sqrt{(D)}$ for $D < 0$ is expressed using

Options

- A. $i = \sqrt{(-1)}$
- B. w
- C. π
- D. e

Correct answer: $i = \sqrt{(-1)}$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $i = \sqrt{(-1)}$.